Macro-systemic Approach to Multi-Asset Investing

Risk Profile Analysis: US 20+ Year Treasury Bond (TLT)

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Buy-Side

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Executive summary

This document is part of a macro-driven multi-asset allocation research project. The objective is to design a rules-based framework that translates macroeconomic regimes into systematic portfolio allocations and decisions across a set of assets, targeting risk-adjusted alpha over equity benchmarks.

This section covers individual asset analytics: return distribution, performance, risk, volatility, Sharpe, Sortino, VaR, CVaR, maximum drawdown, and time under water. Each asset is analyzed independently before being integrated into the broader cross-asset and allocation framework.

This report focuses on 20Y+ Treasury Bonds (TLT) as they are long-duration instruments acting as a deflation hedge and defensive anchor in growth slowdown regimes.

Selected Assets

Class	Product	Ticker	First Obs.
Equity	ETF S&P 500	SPY	29-01-1993
Commodities	SPDR Gold Shares	GLD	18-11-2004
Commodities	Commodity Index Tracking Fund	DBC	06-02-2006
Bonds	Treasuries 20Y+	TLT	30-07-2002
Credit*	iShares Corp Bond	LQD	TBD
Common period			2006-02-06

Table: Selected assets for a multi-asset portfolio

*To be implemented after further research on the credit cycle and its impact on the economy.

20Y+ Bonds - General description

Characteristics:

- **Largest** and **most liquid** 20+ year Treasury ETF globally
  - \$45+ billion AUM, 0.15% expense ratio
  - Tracks ICE U.S. Treasury 20+ Year Bond Index
- **Holdings composition:** 99.9% of AA US Treasury bonds
  - 45 holdings, average maturity 25.8 years
  - Yield to Maturity: $\sim$4.9%
  - Average coupon: $\sim$3.4%
- **Interest rate sensitivity** (Duration $\approx$ 17 years)
  - *Risk:* $-$17% if 20Y+ Treasury yields rise 1%
  - *Hedge:* +20% during crises (2008, 2020) when yields collapse
- **Portfolio role:** Primary deflation/growth-slowdown hedge
  - Negative equity correlation during recessions $\Rightarrow$ *Growth*$\downarrow$

20Y+ Bonds - Historic performances

The **TLT (iShares 20+ Year Treasury Bond ETF)** serves as a portfolio hedge against growth slowdowns and deflation.

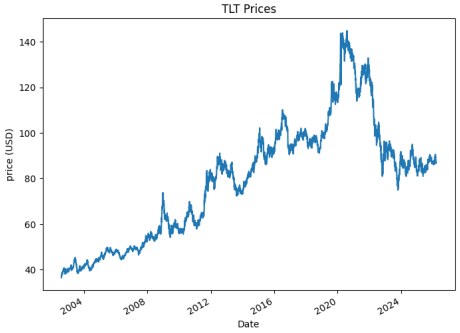


Figure: TLT historical price evolution using market closing prices (ex-coupon)

Performance:

- **Period:** 2002–2026 (24 years)
- **CAGR:** 3.78%
- **Annualized Volatility:** 14.3%
- **Sharpe Ratio:** 0.223
- **Sortino Ratio:** 0.318

Insight: TLT exhibits lower return potential than SPY but with reduced volatility. Low Sharpe confirms bonds as diversifiers, primarily for tail protection.
Sortino > Sharpe reveals favorable asymmetry: losses are offset by gains.

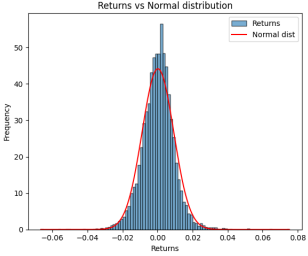
20Y+ Bonds - Distribution of returns

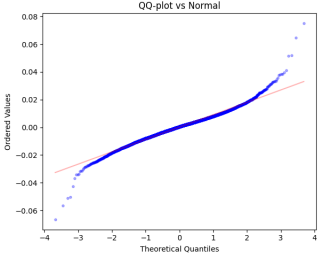
Distribution parameters

Metric	Value
Mean	0.019%
Skewness	0.055
Kurtosis	3.48
Exc. Kurtosis	0.48

Risk metrics

CL	VaR	CVaR
95%	1.46%	1.99%
97.5%	1.80%	2.38%
99%	2.31%	2.92%





- **Small tails** and **Near normal** distribution
- **Q-Q deviation** Slight positive deviation in upper tail ⇒ occasional large gains
- **Near-zero skew** indicates symmetric return distribution: losses and gains are equally likely around the mean.

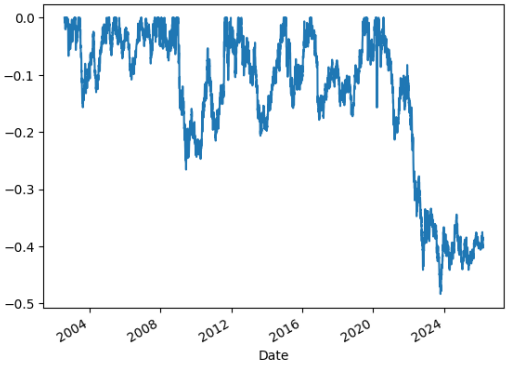
Insights:

→ Near-normal distribution and controlled tail risk provide reliable equity protection versus SPY's extreme drawdowns in credit panic regimes, as Slight upper tail convexity captures outperformance during flight-to-quality events. But it stays unreliable in inflation-driven rate shocks

→ Key portfolio edge: Stable hedge behavior justifies systematic overweight in defensive regimes.

20Y+ Bonds - Drawdowns analysis

Historical drawdowns:



Insights: TLT typically exhibits moderate drawdowns with fast recovery compared to SPY during heavy macro shocks.

- **Examples**: 2012 taper tantrum (-20.7%, 11 months), 2015 oil/China crisis (-15.8%, 8 months).
- **Exception**: Ongoing drawdown since Aug 2020 peak (-48.4%, 1410 periods = 3.9 years) caused by the post-COVID inflation shock $\Rightarrow$ Fed hikes crushed long yields from 1.2% to 4.6%.
  This exception contrasts with the 2008/2020 crises, showing high regime dependency.

$\rightarrow$ Reduce TLT exposure in Policy Tight + Inflation environments where hedge fails.

20Y+ Bonds - Drawdowns analysis

Longest & deepest Dd – IR increase (2020)

Metric	Value
Peak	Aug 4, 2020
Trough	Oct 19, 2023
Recovery	Ongoing (TBD)
Max Drawdown	**-48.35%**
Time Under Water	1,410+ days
Recovery Time	Still ongoing

Taper Tantrum – FED signals end of QE (2012)

Metric	Value
Peak	Jul 26, 2012
Trough	Aug 21, 2013
Recovery	Dec 10, 2014
Max Drawdown	**-20.66%**
Time Under Water	867 days
Recovery Time	477 days

Global Financial Crisis (2008–2011)

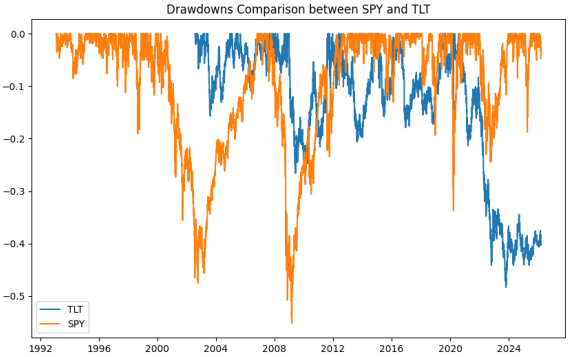
Metric	Value
Peak	Dec 30, 2008
Trough	Jun 10, 2009
Recovery	Aug 18, 2011
Max Drawdown	**-26.59%**
Time Under Water	664 days
Recovery Time	553 days

Oil price decrease & China Crisis (2015)

Metric	Value
Peak	Feb 2, 2015
Trough	Jun 26, 2015
Recovery	Jun 9, 2016
Max Drawdown	**-15.81%**
Time Under Water	493 days
Recovery Time	348 days

20Y+ Bonds - Comparison with SPY

Historical drawdowns:



2008 GFC : TLT vs SPY Comparison

Metric	TLT	SPY
Peak	Dec 30, 2008	Oct 9, 2007
Trough	Jun 10, 2009	Mar 9, 2009
Recovery	Aug 8, 2011	Aug 16, 2012
Max Drawdown	-26.59%	-55.19%
TUW	664d	1,224d
Recovery	553d	869d

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Macro-systemic Approach to multi-asset investing

Nicolas Poncin is a M.Sc. Financial Engineering student with a strong interest in global macroeconomics and multi-asset portfolio management. He aims to apply his academic background to real investment decision-making. This project was developed as a practical research initiative to explore how macroeconomic regimes influence asset behaviour and portfolio allocation, intending to build a disciplined and testable investment process.



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